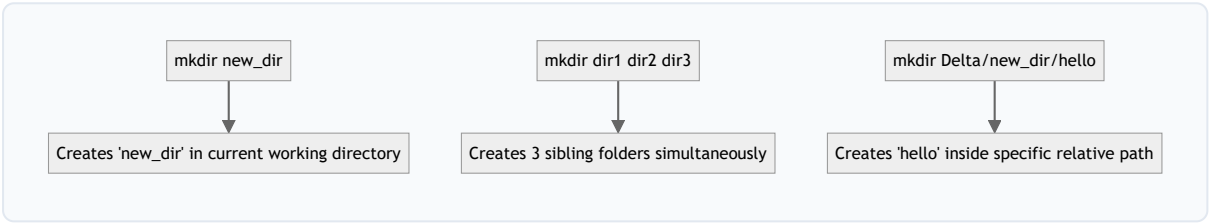
Submodule 34.08: Directory Creation: `mkdir`, Nested Trees, and Path Arguments

Topic Overview & Syntax

The `mkdir` **(Make Directory)** command is the universal shell utility for creating new folders. Instead of right-clicking and navigating GUI dialogs, `mkdir` instantiates single directories, multiple sibling directories, or nested folder paths in a single command.



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1. Core `mkdir` Operations

A. Creating a Single Directory

To create a new directory inside your current working directory:

```
mkdir new_directory
```

Verify creation using `ls` to observe `new_directory` in the output listing.

B. Creating Multiple Directories Simultaneously

Pass space-separated directory names to create multiple directories in a single command:

```
mkdir frontend backend database
```

This instantly generates all three directories.

C. Creating Directories via Relative and Absolute Paths

You do not need to `cd` into a directory to create a subfolder inside it. Supply the relative or absolute path as the argument:

```
# Using a relative path
mkdir Delta/new_dir/hello

# Using an absolute path
mkdir /Users/Shraddha/Desktop/Delta/new_dir2/hello2
```

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2. Shell Rules & Error Behaviors

A. Collision Error (Directory Already Exists)

If a directory with the specified name already exists in that location, the shell refuses to overwrite it:

```
mkdir hello
# mkdir: hello: File exists
```

Takeaway: The terminal ensures that creating a directory never inadvertently destroys or overwrites an existing folder.

B. Missing Intermediate Directories

If intermediate parent directories do not yet exist, `mkdir` throws an error:

```
mkdir parent/child/grandchild
# mkdir: parent/child: No such file or directory
```

Pro Tip (Flag `-p`): Using the parent flag `mkdir -p parent/child/grandchild` automatically generates all missing intermediate parent directories along the path.

C. Handling Folder Names with Spaces

If a directory name contains whitespace, the shell treats each word as a separate folder:

- `mkdir my new folder` → Creates 3 distinct folders: `my`, `new`, and `folder`!
- **Solution (Quotes or Escape Character)**:

```
mkdir "my new folder"     # Enclosed in quotes
mkdir my\ new\ folder     # Backslash space escaping
```